

CYBOX-S: OPERATOR GUIDE & PRACTICAL WORKFLOW REFERENCE

High-Speed Offline Penetration Testing Worksheet · eJPTv2 / eCPPT Practical Companion

What is CYBOX-S?

CYBOX-S is a fast, keyboard-driven terminal worksheet and operational cockpit designed to eliminate exam cognitive overload. It provides an offline state machine for host discovery, port tracking, credential reuse, syntax cheatsheets, and question proofs. **It does not run autonomous exploits or rely on AI.** You retain 100% human control while CYBOX-S manages your operational memory.

INE eJPT / eCPPT Exam Compliance

- **100% Passive & Offline:** Runs entirely on localhost via local SQLite database.
- **Zero Autonomous Action:** Commands are copied to your clipboard; YOU execute them.
- **Zero Cloud / AI Dependencies:** No external API calls, leaks, or prohibited LLMs.
- **Permitted Personal Notes:** Functions strictly as a local worksheet and syntax lookup.

1. The Four Operational Stations (Switch with Keys 1, 2, 3, 4)

Station	Name & Purpose	What You See & Do Here	Hotkey
Station 1	Cockpit (Workbench) Primary attack surface station	Targets tree, open ports, service status triage, methodology checklist, field notes, and interactive command console.	[1]
Station 2	Playbook Browser Methodology & cheatsheet search	Browse full tactical playbooks (eJPT workflow, Web App OWASP, Active Directory, Pivoting, Linux & Windows PrivEsc). Search commands with instant copy.	[2]
Station 3	Credential Matrix 2D Vault & Lateral Spray Grid	2D grid of discovered credentials (rows) × target services (cols). Press [Enter] to compile & copy spray commands; press [Space] to cycle verification state.	[3]
Station 4	Exam Proofs & Loot Q1–Q35 ledger & flags	Track exam question answers (Q1–Q35 via :q or [a]), user/root flags, and Failure Logs . Run :export exam to generate an offline Markdown submission report.	[4]

2. Station 1 Anatomy & Screen Real Estate

Cockpit is mathematically arranged in a zero-overflow 4-quadrant layout with a dedicated sidebar and bottom console bar:

SIDEBAR (Left 28% width) • ATTACK SURFACE: Collapsible tree of target IPs and hostnames. Displays live service badges. • CREDENTIALS PREVIEW: Discovered usernames and masked passwords. Press [Space] to reveal. <i>Tip: Press [b] anytime to toggle/collapse sidebar and give 100% width to Workbench!</i>	WORKBENCH (Right 72% width) • SERVICES & PORTS (Top): Port, protocol, version, and potential. Press [Space] to cycle triage status. Highlight service to auto-filter checklist! • METHODOLOGY (Lower-Left): Step-by-step checklist with progress bar. Auto-tracks current port. • NOTES & FINDINGS (Lower-Right): Live scratchpad for leads, vulns, and flags.
BOTTOM CONSOLE BAR (Full Width) — Fast-capture command line & live syntax guidance drawer. • Top Line: Ready-to-paste command with target IP pre-substituted. Press [Enter] or [y] to copy. • Second Line: Tactical explanation and parameter tip. • Input Row: Press [:] to focus. Type natural notes, commands (e.g. :w rockyou), or capture data.	TOP MACHINE STATUS STRIP — Instant context on current active target. • Shows Target IP, Hostname, OS, Scope badge ([IN-SCOPE]). • Live progress bar showing methodology completion percentage. • NEXT ACTION recommendation highlighting the immediate next step.

3. Complete Keyboard Shortcuts Cheatsheet ("Muscle Memory Map")

CYBOX-S is 100% operational from the keyboard. Single-letter keys trigger immediate actions unless an input dialog is active.

Key	Category	Action & Operational Behavior
1, 2, 3, 4	Station	Switch immediately to Station 1 (Cockpit), 2 (Playbooks), 3 (Creds Matrix), or 4 (Loot).
w	Navigation	Cycle Panels: Sequentially rotates focus across all 5 panels in Cockpit without using Tab.
h / l	Navigation	Column Jump: Press h to jump to Sidebar (left); press l to jump to Workbench (right).
j / k	Navigation	Vim Navigation: Move highlight down (j) or up (k) in the active list or tree.
[/]	Targeting	Target Switcher: Switch immediately to previous [or next] target machine in your workspace.
b	Display	Sidebar Toggle: Collapse sidebar to give 100% horizontal width to Workbench (press b again to restore).
: (colon)	Console	Focus Command Bar: Instantly activates the bottom console input ready to type.
Esc	Console	Dismiss / Blur: Clears the command input and returns cursor focus directly to the workbench list.
Space	Triage / Reveal	• On Service: Cycle status: UNTESTED → [CHECKED] → [DEAD-END] → [DEFERRED]. • On Credential: Reveal/mask secret password. • In Station 3 Matrix: Cycle test state: [UNTESTED] → [VALID] → [PWN3D] → [INVALID].
, / .	Tool Carousel	Multi-Tool Recipe Carousel: When a service is highlighted, press . (next) or , (prev) to cycle alternative tool recipes in the console (e.g. feroxbuster → gobuster → nikto → curl).
Enter	Execute / Copy	• On Guidance / Service: Copies the previewed tool command to system clipboard. • In Station 3 Matrix: Compiles and copies ready-to-run spray command for highlighted credential & port.
y	Copy	Quick Copy: Copies the selected entity's primary value (IP address, port, password, or checklist command).
z	Layout	Zoom: Maximize the focused panel to full-screen; press z again to restore 4-quadrant layout.
T	Theme	Theme Picker: Open visual theme modal with 10 vibrant palettes (press d to set default).
/ or Ctrl+F	Search	Global Fuzzy Search: Search across targets, services, credentials, checklists, and notes.
r	Reference	Cheat Sheet: Open offline practical reference dialog for quick syntax inspection.
t	Fast Capture	Open Add Target dialog (or type :t <ip> [host] [os] in console).
s	Fast Capture	Open Add Service dialog (or type :s <port/proto> <service> in console).
c	Fast Capture	Open Add Credential dialog (or type :c <user:pass> [scope] in console).
n	Fast Capture	Open Add Note dialog (or type :n <text> in console).
f	Fast Capture	Open Add Finding / Vulnerability dialog (or type :f <title> in console).
m	Methodology	Open Methodology Picker dialog (or type :m <name> in console).
a	Exam Proof	In Station 4: Open Add Exam Proof modal to link question number (Q1–Q35) to proof evidence.
d	Manage	Delete highlighted entity (requires confirmation).

PRO-TIP: THE ZERO-MOUSE OPERATOR FLOW

1. Highlight any port in **SERVICES & PORTS** with j / k.
2. Watch the bottom console display the exact syntax with the target IP already substituted.
3. Press . to cycle alternative tools (e.g., feroxbuster → gobuster → nikto → curl).
4. Press Enter to copy the command directly to your terminal clipboard, switch to your tmux/terminal pane, and run it!
5. Press Space to mark the port [CHECKED] or [DEAD-END].

4. Bottom Console Syntax & Wordlist Accelerators

Console Command	Description & Operational Effect	Example Syntax
<code>:w <alias></code>	Wordlist Accelerator: Instantly copies standard Kali / SecLists paths to clipboard.	<code>:w rockyou</code> <code>:w common</code> <code>:w medium</code> <code>:w raft-d</code> <code>:w raft-f</code> <code>:w users</code>
<code>:t <ip> [opts]</code>	Add target with optional host, OS, subnet, and pivot flag.	<code>:t 10.10.10.5 host linux</code> <code>10.10.10.0/24 pivot</code>
<code>:pivot / :subnet</code>	Tag pivot machine / assign subnet CIDR for Station 1 tree grouping.	<code>:pivot 10.10.10.5 172.16.1.0/24</code> <code>:subnet 10.10.10.5 10.10.10.0/24</code>
<code>:s <port> <svc></code>	Record newly discovered port & service on active target.	<code>:s 445/tcp smb</code> <code>:s 8080 http</code>
<code>:c <user:pass> [scope]</code>	Record discovered credential with optional service scope.	<code>:c admin:Secret123! SMB</code> <code>:c root:toor SSH</code>
<code>:q / :export</code>	Pin exam question proof (Q1–Q35) or export markdown exam dossier.	<code>:q 7 /var/www/wp-config.php</code> <code>:export exam</code> <code>:export report</code>
<code>:uflag / :rflag <hash></code>	Record user flag or root/admin flag into Station 4 Loot Ledger.	<code>:uflag 7a9e...</code> <code>:rflag f04b...</code>
<code>:st <where> / :cl <clue></code>	Record stuck point (rabbit hole) and breakthrough clue.	<code>:st stuck on wp-login</code> <code>:cl found backup.zip</code>
<code>:m <template></code>	Instantiate standard checklist template (replace or append).	<code>:m ejpt</code> <code>:m web</code> <code>:m smb append</code>
<code>:theme <name></code>	Switch theme live (e.g. sugary, caramel, midnight, cyber).	<code>:theme sugary</code> <code>:theme midnight</code>

5. Step-by-Step Practical Exam Workflow ("The Battle Plan")

Follow this methodical 5-phase execution cycle to maintain relentless momentum and pass calmly:

PHASE 1: SCOPING & HOST DISCOVERY 1. Host Discovery: Identify alive target IPs (nmap -sn 10.10.10.0/24 or arp-scan). 2. Fast Add to CYBOX-S: Type <code>:t 10.10.10.20</code> into console. Assign subnets: <code>:subnet 10.10.10.20 10.10.10.0/24</code>. 3. Scope & Routing: Tag dual-homed pivot machines with <code>:pivot 10.10.10.20 172.16.1.0/24</code> for automatic guidance. 4. Import XML Option: Mass scans can be imported directly: <code>cybox-s import /path/to/scan.xml</code>.
PHASE 2: DETAILED PORT & SERVICE ENUMERATION 1. Full Port Sweep: <code>nmap -sS -p- --min-rate 1000 <IP></code>, then version scan: <code>nmap -sV -sC -p <ports> <IP></code>. 2. Record Open Services: Type <code>:s 80 http</code>, <code>:s 445 smb</code>, <code>:s 3306 mysql</code> into the console. 3. Inspect Tactical Guidance: Highlight each service in SERVICES & PORTS. Press <code>.</code> to cycle tool recipes (whatweb → feroxbuster → gobuster → nikto → curl). Press <code>[Enter]</code> to copy. 4. Service Triage with [Space]: Cycle status: UNTESTED → [CHECKED] → [DEAD-END] → [DEFERRED].
PHASE 3: WEB & LOW-HANGING FRUIT EXPLOITATION 1. Inspect Anonymous / Default Access: Check anonymous FTP (<code>:s 21</code>), null SMB shares (<code>:s 445</code>), and SNMP strings (<code>:s 161</code>). 2. Directory Fuzzing: Type <code>:w common</code> or <code>:w medium</code> to copy the SecLists directory path. Paste directly into feroxbuster or gobuster. 3. Cross-Filtering: Highlighting a service in SERVICES & PORTS auto-scrolls the Methodology checklist to that service's testing steps!
PHASE 4: CREDENTIAL SPRAYING & LATERAL MOVEMENT (STATION 3) 1. Record Every Credential: Whenever you find creds (web configs, DB dumps, code), press <code>[c]</code> or type <code>:c user:pass scope</code>. 2. Open Station 3 Matrix: Press <code>[3]</code> to switch to the 2D Credential Spray Matrix. 3. Instant Spray Generation: Highlight any cell and press <code>[Enter]</code> to compile & copy ready-to-run spray commands (netexec, evil-winrm). 4. Record Verification State: Press <code>[Space]</code> on the cell to cycle: [UNTESTED] → [VALID] → [PWN3D] → [INVALID]. State persists in SQLite!
PHASE 5: PRIVESC, PIVOTING & EVIDENCE CAPTURE (STATION 4) 1. Pivoting & Route Addition: Tag pivot hosts with <code>:pivot</code> to see dynamic proxy guidance in the console bar. 2. Privilege Escalation: Switch to Station 2 (press <code>[2]</code>) and search Linux SUID, sudo -l, Cron jobs, or Windows tokens / services. 3. Capture Question Evidence & Flags: Log question proofs with <code>:q <num> <proof></code> (or press <code>[a]</code> in Station 4). Record flags with <code>:uflag / :rflag</code>. 4. Export Exam Submission Bundle: Run <code>:export exam</code> to write a clean Markdown report with all Q1–Q35 evidence, flags, and pivot paths!
THE 20-MINUTE ROTATION RULE (EXAM GOLDEN RULE) If you have been analyzing a single service for more than 20 minutes without a new lead: STOP. Press <code>[Space]</code> on the service to mark it [DEFERRED]. Type <code>:stuck <service description></code> to log where you are stuck, press <code>]</code> to rotate to another machine or port, and harvest low-hanging fruit elsewhere. 80% of exam failures occur because students spend 3 hours trapped down an unexploitable rabbit hole!

6. Station 3 Deep Dive: 2D Credential Spray Matrix

In multi-machine exams (like eJPTv2 / eCPPT), credential reuse across SSH, SMB, WinRM, and Web portals is the #1 lateral movement technique. Station 3 prevents credential amnesia by organizing credentials against in-scope authenticating services:

Credential (Row)	10.10.10.20:22 (SSH)	10.10.10.20:445 (SMB)	10.10.10.25:5985 (WinRM)	10.10.10.30:3389 (RDP)
admin : P@ssword123	[VALID]	[PWN3D]	[UNTESTED]	[INVALID]
guest : <empty>	[INVALID]	[VALID]	[INVALID]	[UNTESTED]
dbuser : dbpass2024	[UNTESTED]	[INVALID]	[UNTESTED]	[UNTESTED]

Navigation: Use arrow keys to select any cell. Press **[Enter]** to instantly copy the command (e.g. netexec smb 10.10.10.20 -u admin -p 'P@ssword123'). Press **[Space]** to cycle status: [UNTESTED] → [VALID] → [PWN3D] → [INVALID].

7. Built-In Methodology Templates (Type :m)

Template Alias	Methodology Category	Key Checklist Steps & Focus
:m ejpt	eJPT Practical Pentest	Full 12-step structured workflow: Scope recon → Host discovery → Full TCP scan → Service detection → Low-hanging fruit → Web fuzzing → Auth inspection → Exploitation → Pivoting → PrivEsc.
:m web	OWASP Web Assessment	Technology profiling (whatweb) → Content discovery (feroxbuster/gobuster) → Input parameter fuzzing → Authentication bypass → SQL Injection → File inclusion (LFI/RFI) → Upload forms.
:m smb	SMB & Active Directory	Null session shares (smbclient/smbmap) → User & group enumeration (rpcclient) → Password policy check → Anonymous signing audit → Vulnerability scan (MS17-010, ZeroLogon).
:m pivoting	Network Pivoting & Routing	Dual-homed adapter discovery (ip a) → Internal route table inspect → Proxychains & Chisel server setup → Port forwarding (socat) → Internal segment host sweep.
:m privesc_linux	Linux Local PrivEsc	sudo -l commands → SUID/SGID binaries (GTF0Bins) → Capabilities (getcap) → World-writable scripts & crontabs → Internal listening ports → Password files (/etc/shadow permissions).
:m privesc_windows	Windows Local PrivEsc	whoami /priv (SeImpersonate) → Unquoted service paths → Modifiable service binaries → Stored credentials (cmdkey /list) → AlwaysInstallElevated → SAM/SYSTEM registry backup.

8. Exam-Day Readiness & CLI Reference

EXAM START PROCEDURES:

1. Open terminal on your host or VM and start: cyb0x-s

2. Pick your favorite visual theme: press **[T]**, select palette (e.g. *Sugary*, *Midnight*, *Slate*, or *Cyber*), press **[d]** to save as default.

3. Import initial Nmap scan: type cyb0x-s import /path/to/scan.xml or fast-add targets with :t <ip> (with optional subnet & pivot tags).

4. **Offline Markdown Submission Bundle:** Run :export exam at any time to output a structured markdown report containing all Q1–Q35 evidence, flags, credentials, and pivot paths.

5. **Database Safety:** Stored in ~/.local/share/cyb0x-s/worksheets.db. 100% local, persistent, zero network calls, and fully compliant with INE exam rules.